```python
  This function calculates the nth Factorial number using recursion.


def factorial():
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)


number = int(input("Enter value of n: "))
result = factorial(number)
print(f"The factorial of {number} is: {result}")
```

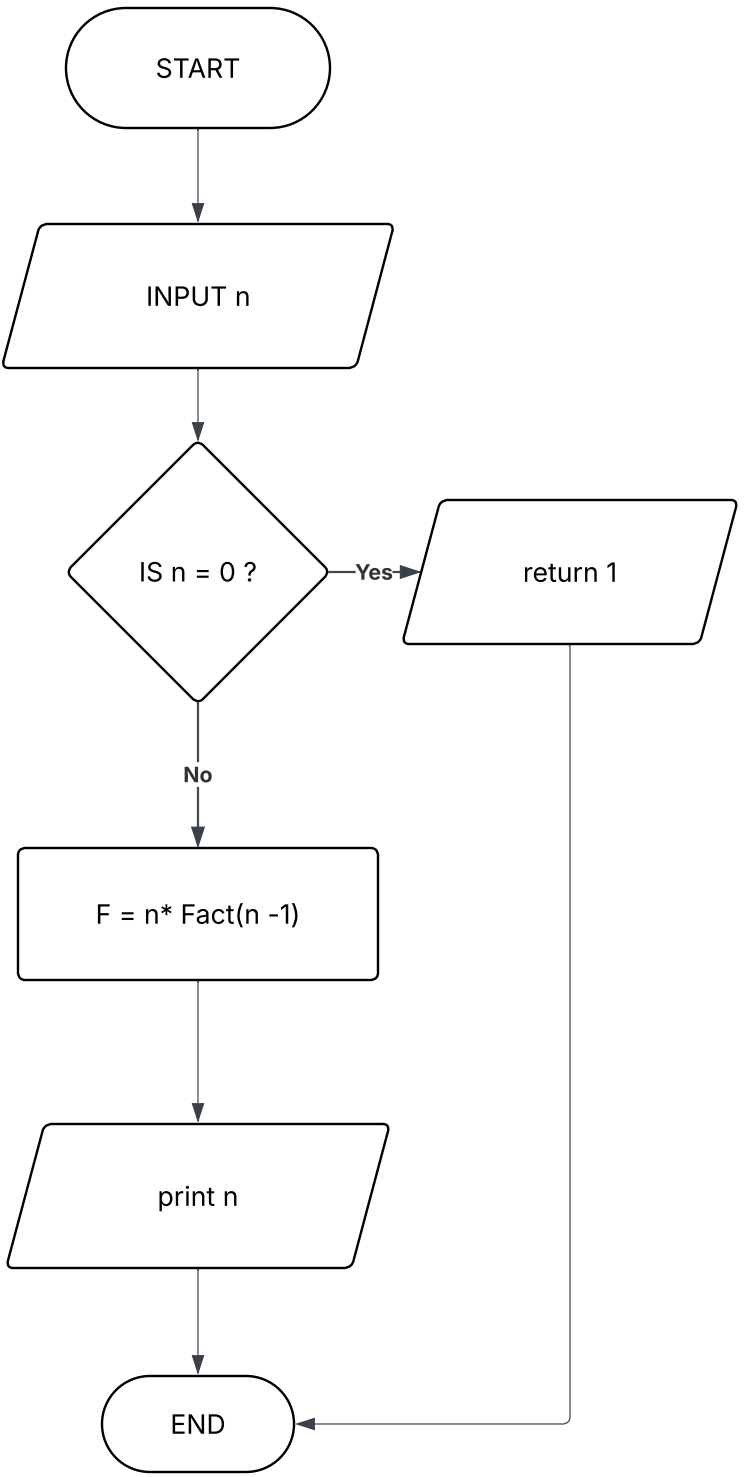
PSEUDOCODE

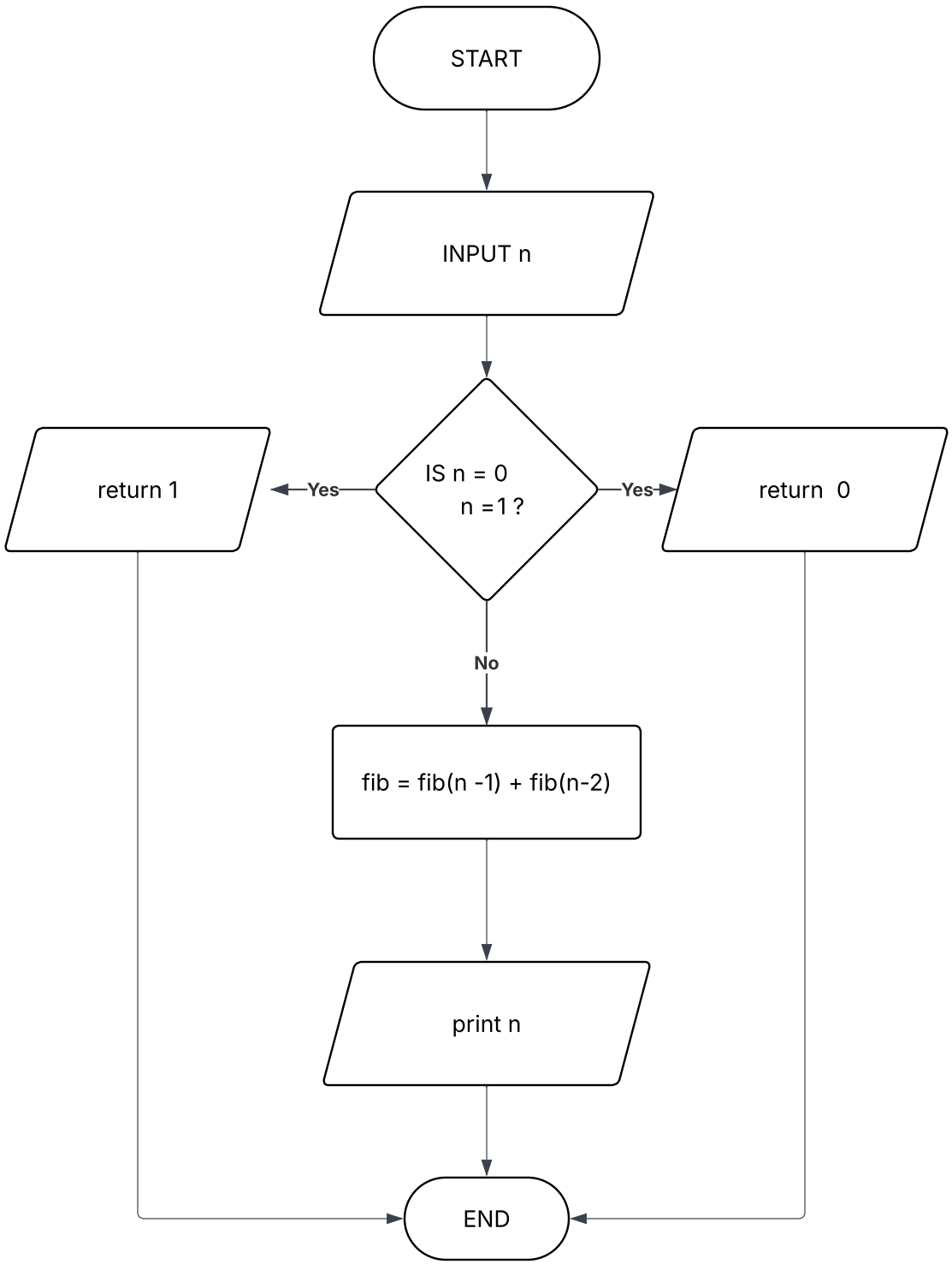
```
start
  function factorial(n)
    enter n
    if n == 0
      return 1
    else
       return n * factorial(n-1)
    end if
   set result = factorial(n)
   display result
stop
```

```python
This function calculates the nth Fibonacci number using recursion.

def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

#ACCEPT USER INPUT
number = int(input("Enter the value of n: "))
result = fibonacci(number)
print(f"The {number}th Fibonacci number is: {result}")
```

PSEUDOCODE

```
start
  function fibonacci(n)
    enter n
    if n == 0
      return 1
    else if n == 1
      return 1
    else
       return fib(n-1) + fib(n-2)
    end if
   set result = fibonacci(n)
   display result
stop
```


START
INPUT n
IS n = 0 ?
Yes
return 1
No
F = n* Fact(n -1)
print n
END


START
INPUT n
IS n = 0
n =1 ?
Yes
return 1
Yes
return 0
No
fib = fib(n -1) + fib(n-2)
print n
END